

Table 1: Survey of Applied Papers

Paper	Binary Absorbing Treatment	Variation in Treatment Timing	Pre-Treatment Relative Period Excluded	Exclude Distant Relative Periods	Bin Distant Relative Periods	Includes Never Treated Units	Panel Balanced in Relative Time
Bosch and Campos-Vazquez (2014)	X						
Fitzpatrick and Lovenheim (2014)							
Gallagher (2014)	X	X	-1		X	X	
Tewari (2014)	X	X	0				X
Ujhelyi (2014)	X	X	-1		X	X	
Bailey and Goodman-Bacon (2015)	X	X	-1		X	X	X
Deryugina (2017)	X	X	-1		X	X	X
Deschenes et al. (2017)	X						
He and Wang (2017)	X	X	-1		X		
Lafortune et al. (2017)	X	X	0	X		X	
Kuziemko et al. (2018)	X	X	-1		X		
Markevich and Zhuravskaya (2018)							

Notes: This table consolidates key properties of main event study specifications across a sample of applied papers. We follow the selection criteria in Roth (2019): the original sample consists of 70 total papers, but is further constrained to these twelve papers with publicly available data and code. The data and code are used to determine exactly the specification estimated in these papers. We focus on the first specification underlying the event study estimates in each paper, which we view as a reasonable proxy for the main specification in the paper. Note that two papers (Fitzpatrick and Lovenheim (2014); Markevich and Zhuravskaya (2018)) have none of the attributes listed in the columns.

Table 2: Summary Statistics of the HRS Sample

	<i>N</i>	Mean	Std. Dev
<i>Panel A. Demographics</i>			
Age at admission	656	56	2.29
Male	656	0.456	0.498
Year of admission	656	2,007	2.11
<i>Panel B. Race/ethnicity</i>			
Hispanic	656	0.122	0.327
Black	656	0.151	0.358
White	656	0.742	0.438
Other race	656	0.107	0.309
<i>Panel C. Insurance status</i>			
Medicaid	656	0.05	0.219
Private	656	0.715	0.452
Medicare	656	0.072	0.259
<i>Panel D. Pre-hospitalization outcome</i>			
Out-of-pocket medical spending			
Wave 7 ($E_i \geq 1$)	656	3,302	9,024
Wave 8 ($E_i \geq 2$)	404	2,355	8,132
Wave 9 ($E_i \geq 3$)	228	2,056	3,532
Wave 10 ($E_i = 4$)	65	2,044	4,379
Earnings			
Wave 7 ($E_i \geq 1$)	656	43,810	67,950
Wave 8 ($E_i \geq 2$)	404	38,944	58,601
Wave 9 ($E_i \geq 3$)	228	36,274	56,768
Wave 10 ($E_i = 4$)	65	29,037	46,289

Notes: This table presents summary statistics on our primary analysis sample, taken from the biennial Health and Retirement Survey (HRS). We include the sample of individuals ages 50-59 in waves 7-11 (approximately spanning 2004-2012) who appear in two sequential survey waves and report a recent hospital admission in the second survey. For panel D, the sample corresponding to wave t is conditional on not having hospitalization by wave t .

Table 3: Estimates for the Effect of Hospitalization on Outcomes

(a) Out-of-pocket Medical Spending					
ℓ Wave Relative to Hospitalization	FE $\hat{\mu}_\ell$	IW $\hat{\nu}_\ell$	$\hat{\delta}_{1,\ell}$	$CATT_{e,\ell}$ $\hat{\delta}_{2,\ell}$	$\hat{\delta}_{3,\ell}$
-3	149 (792)	591 (1273)	-	-	591 (1273)
-2	203 (480)	353 (698)	-	299 (967)	411 (1030)
-1	0	0	0	0	0
0	3,013 (511)	2,960 (543)	2,826 (1038)	3,031 (704)	3,092 (998)
1	888 (664)	530 (587)	825 (912)	107 (653)	-
2	1,172 (983)	800 (1010)	800 (1010)	-	-
3	1,914 (1426)	-	-	-	-
(b) Labor Earnings					
ℓ Wave Relative to Hospitalization	FE $\hat{\mu}_\ell$	IW $\hat{\nu}_\ell$	$\hat{\delta}_{1,\ell}$	$CATT_{e,\ell}$ $\hat{\delta}_{2,\ell}$	$\hat{\delta}_{3,\ell}$
-3	-2,642 (3504)	-8,228 (6592)	-	-	-8,228 (6592)
-2	-5,089 (3005)	-7,823 (4197)	-	-7,691 (6348)	-7,964 (4766)
-1	0	0	0	0	0
0	-1,225 (2742)	-863 (2967)	455 (5593)	3,032 (4152)	-7,107 (5883)
1	-7,508 (4312)	-5,435 (4348)	-1,670 (6500)	-10,826 (4863)	-
2	-11,102 (5976)	-10,670 (6154)	-10,670 (6154)	-	-
3	-8,780 (8332)	-	-	-	-

Notes: This table reports three different sets of estimates for the dynamic effects of hospitalization on out-of-pocket medical spending in panel (a) and labor earnings in panel (b). The first column reports estimates from the FE estimator $\hat{\mu}_\ell$. The sample includes observations from wave $t = 0, 1, 2, 3, 4$. Columns 3-5 report the estimates for $CATT_{e,\ell}$ from $\hat{\delta}_{e,\ell}$. The sample includes observations from wave $t = 0, 1, 2, 3$. Column 2 reports the IW estimates which are constructed as the weighted average of $\hat{\delta}_{e,\ell}$'s across cohorts e who are ℓ periods from hospitalization. Standard errors (clustered on the individual) are shown in parentheses.

A Notation glossary and properties of double-demeaning

In this section, we provide relevant notations for proofs in Section B. Section A.1 introduces detailed expression for the weight.

We observe a balanced panel of N i.i.d. observations $\{\{Y_{i,t}, D_{i,t}\}_{t=0}^T\}_{i=1}^N$ where $Y_{i,t} \in \mathbb{R}$ is a real valued outcome variable and $D_{i,t} \in \{0, 1\}$ is a binary absorbing treatment status variable: $D_{i,t} = 0$ if i is not treated in period t and $D_{i,t} = 1$ if i is treated in period t .

Since the treatment is absorbing, we can aggregate the treatment path $\{D_{i,t}\}_{t=0}^T$ to a single discrete random variable $E_i = \min\{t : D_{i,t} = 1\}$, which is the period of the initial treatment. Additionally, we define $D_{i,t}^\ell := \mathbf{1}\{t - E_i = \ell\}$ to be an indicator for being ℓ time periods relative to unit i 's initial treatment.

We focus on the following two-way fixed effects regression

$$Y_{i,t} = \alpha_i + \lambda_t + \sum_{g \in \mathcal{G}} \mu_g \mathbf{1}\{t - E_i \in g\} + u_{i,t}$$

where g are disjoint sets of relative times, α_i and λ_t are unit and time fixed effects.

Definition 3. For fixed T , consider a random vector $\{X_{i,t}\}_{t=0}^T \in \mathbb{R}^{T+1}$, jointly distributed according to P . At each t , let $\ddot{X}_{i,t}$ denote the following random variable:

$$\ddot{X}_{i,t} = X_{i,t} - E[X_{i,t}] - \frac{1}{T+1} \sum_{s=0}^T X_{i,s} + \frac{1}{T+1} \sum_{s=0}^T E[X_{i,s}] \quad (32)$$

The expectation $E[X_{i,t}]$ is taken cross-sectionally with respect to P at time $t = 0, \dots, T$.

Remark 1. Conventionally $\ddot{X}_{i,t}$ is referred to as the double-demeaned version of $X_{i,t}$, as it removes the contemporaneous expectation $E[X_{i,t}]$ as well as the simple average across time for unit i , $\frac{1}{T+1} \sum_{s=0}^T X_{i,s}$. It adds back a simple average across time of the expectations so that $\ddot{X}_{i,t}$ has the zero-mean and zero-sum properties as detailed in Lemma 1.

Lemma 1. (Properties of double-demeaning.) *For any $\ddot{X}_{i,t}$ and $\ddot{Z}_{i,t}$, double-demeaned versions of $X_{i,t}$ and $Z_{i,t}$ respectively, we have the following properties:*

- zero-mean and zero-sum: $E[\ddot{X}_{i,t}] = 0$ and $\sum_{t=0}^T \ddot{X}_{i,t} = 0$;
- idempotent: $\sum_{t=0}^T E[\ddot{X}_{i,t} \ddot{Z}_{i,t}] = \sum_{t=0}^T E[X_{i,t} \ddot{Z}_{i,t}] = \sum_{t=0}^T E[\ddot{X}_{i,t} Z_{i,t}]$;